

NETWORKS AND COMPLEXITY

Exercise Sheet 20: Motifs

*This is an exercise sheet from the forthcoming book *Networks and Complexity*.
Find more exercises and solutions at <https://github.com/NC-Book/NCB>*

Ex 20.1: Multi-step adjacency [Lvl. 1]

Let λ_n be the eigenvalues of $\mathbf{A}$. Show that the eigenvalues of $\mathbf{A}^k$ are λ_n^k .

Ex 20.2: Zero power [Lvl. 1]

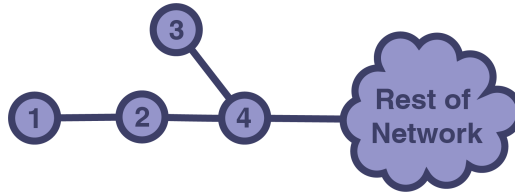
Consider $\sum \lambda_n^k$, where λ_n were the eigenvalues of the adjacency matrix $\mathbf{A}$. This sum reveals interesting network properties for several values of k . Is this also true for $k = 0$?

Ex 20.3: 5-chain [Lvl. 2]

Find the eigenvalues of the adjacency matrix of a chain of five nodes.

Ex 20.4: From the tip of the tail [Lvl. 2]

Consider the following network:



- Suppose the adjacency matrix has an eigenvector $\mathbf{v}$ with eigenvalue λ . Normalize the first element of this eigenvector to $v_1 = 1$, then determine the eigenvector elements v_2 , v_3 and v_4 . For which values of λ is $v_3 > 0$?
- Consider also the case where instead of the branch a chain of 10 nodes is attached to the rest of the network. Compute the eigenvector elements for the nodes in the chain for the case $\lambda = 9$.

Ex 20.5: Epidemic example revisited [Lvl. 3]

We revisit the epidemic model from Ex. 19.8. In this model the Jacobian of the disease-free state is $\mathbf{J} = p\mathbf{A} - r\mathbf{I}$, where p is the infection rate, r is the recovery rate $\mathbf{A}$ is the adjacency matrix and $\mathbf{I}$ is an identity matrix.

- Find the relationship between Jacobian eigenvalues κ_n and the adjacency eigenvalues λ_n .
- You notice that the network contains a fully-connected clique of six nodes. What does this tell you about the largest Jacobian eigenvalue κ_1 ?
- At which value of p will the clique destabilize the disease-free steady state? Is the transition that occurs at this point the epidemic threshold?

Ex 20.6: Competitive exclusion [Lvl. 3]

We consider an exploitative-competition (EC) motif in a food web: Our model includes two populations of predators, X_1 , X_2 , and a resource R , who are part of a large network that

is modeled with many additional variables. The dynamics of our three focal variables are described by

$$\begin{aligned}\dot{X}_1 &= g_1(R)X_1 - m_1X_1 \\ \dot{X}_2 &= g_2(R)X_2 - m_2X_2 \\ \dot{R} &= s(R) - g_1(R)X_1 - g_2(R)X_2 + \dots\end{aligned}$$

where g_1 , g_2 and s are arbitrary functions, m_1 and m_2 are arbitrary mortality rates for the two predator species, and ‘...’ represents further terms that connect R to other variables.

- Formally compute the 3×3 -block of the Jacobian that describes the interaction between X_1 , X_2 , and R . Use the stationarity condition for X_1 and X_2 to simplify your matrix.
- Show that in any large ODE system that contains the EC motif studied here, one eigenvalue of the Jacobian will be zero.

Ex 20.7: Small eigenvalue puzzles [Lvl. 3]

Find the adjacency eigenvalues of the following networks:

Ex 20.8: The star and the snowflake [Lvl. 3]

Consider a node of degree 100, whose neighbors have degree 1.

- Find the eigenvalues and eigenvectors of the network’s adjacency matrix.
- Bonus: Take the 100 nodes of degree 1 and attach two nodes of degree 1 to each of them. Find all eigenvalues and eigenvectors.

Ex 20.9: Bipartite networks [Lvl. 4]

In this exercise we show that bipartite networks have a symmetric spectrum:

- Draw a bipartite simple graph with at least 5 nodes. Color the nodes with two colors such that there are no adjacent nodes with the same color.
- Now number the nodes starting with all nodes of the first color, followed by all nodes of the second color. Show that the adjacency matrix can be written as

$$\mathbf{A} = \begin{pmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{B}^\dagger & \mathbf{0} \end{pmatrix}$$

where $\mathbf{B}$ is a rectangular matrix.

- Assume we know an eigenvector $\mathbf{v}$ of $\mathbf{A}$ with eigenvalue λ . Write $\mathbf{v}$ as two shorter vectors $\mathbf{s}, \mathbf{r}$ stacked on top of each other:

$$\mathbf{v} = \begin{pmatrix} \mathbf{r} \\ \mathbf{s} \end{pmatrix}$$

Show that stacking $\mathbf{r}$ on $-\mathbf{s}$ yields an eigenvector with eigenvalue $-\lambda$.

- Now, prove that every bipartite network has a symmetric spectrum.

Ex 20.10: Reactivity [Lvl. 4]

Let J be the Jacobian of a steady state in an ODE system.

- Consider the deviation $\delta(t)$ from the steady state following an initial perturbation δ_0 , and write an equation of the length $|\delta| = \sqrt{\delta^T \delta}$.

- b) Define a as the relative rate of change in $|\delta|$, i.e. the current change in the distance normalized by the current distance. Then show

$$a = \frac{\delta^\dagger \mathbf{H} \delta}{\delta^\dagger \delta},$$

where $\mathbf{H} = (\mathbf{J} + \mathbf{J}^\dagger)/2$ is the hermitian part of the Jacobian.

- c) Define the reactivity of the steady state as the highest possible value of a . Show that the reactivity is identical to the rightmost eigenvalue of $\mathbf{H}$.
- d) Consider the predator prey system

$$\begin{aligned}\dot{X} &= X - XY/(3 + X) \\ \dot{Y} &= 2XY/(3 + X) - Y.\end{aligned}$$

Compute $\mathbf{J}$ and $\mathbf{H}$ and the reactivity in the nontrivial steady state.

- e) We now consider a larger system where the Jacobian $\mathbf{J}$ from (d) appears as a principal submatrix. (Perhaps the system contains X and Y along with many other variables). What can you say about the reactivity of the entire system?

Ex 20.11: More proofs [Lvl. 4]

Prove the following using direct computation, Gershgorin's theorem or Rayleigh quotient maximization, or eigenvalue properties:

- Regular graphs have an eigenvalue that is identical to their mean degree.
- The highest node degree is an upper bound for the largest eigenvalue in any network.
- The mean degree is a lower bound for the rightmost adjacency eigenvalue in any undirected simple graph.
- Bipartite networks cannot contain cycles of odd length (prove this one algebraically, don't use the coloring intuition).

Ex 20.12: A large eigenvalue puzzle [Lvl. 5]

Find as many eigenvalues as possible of the dragon network: